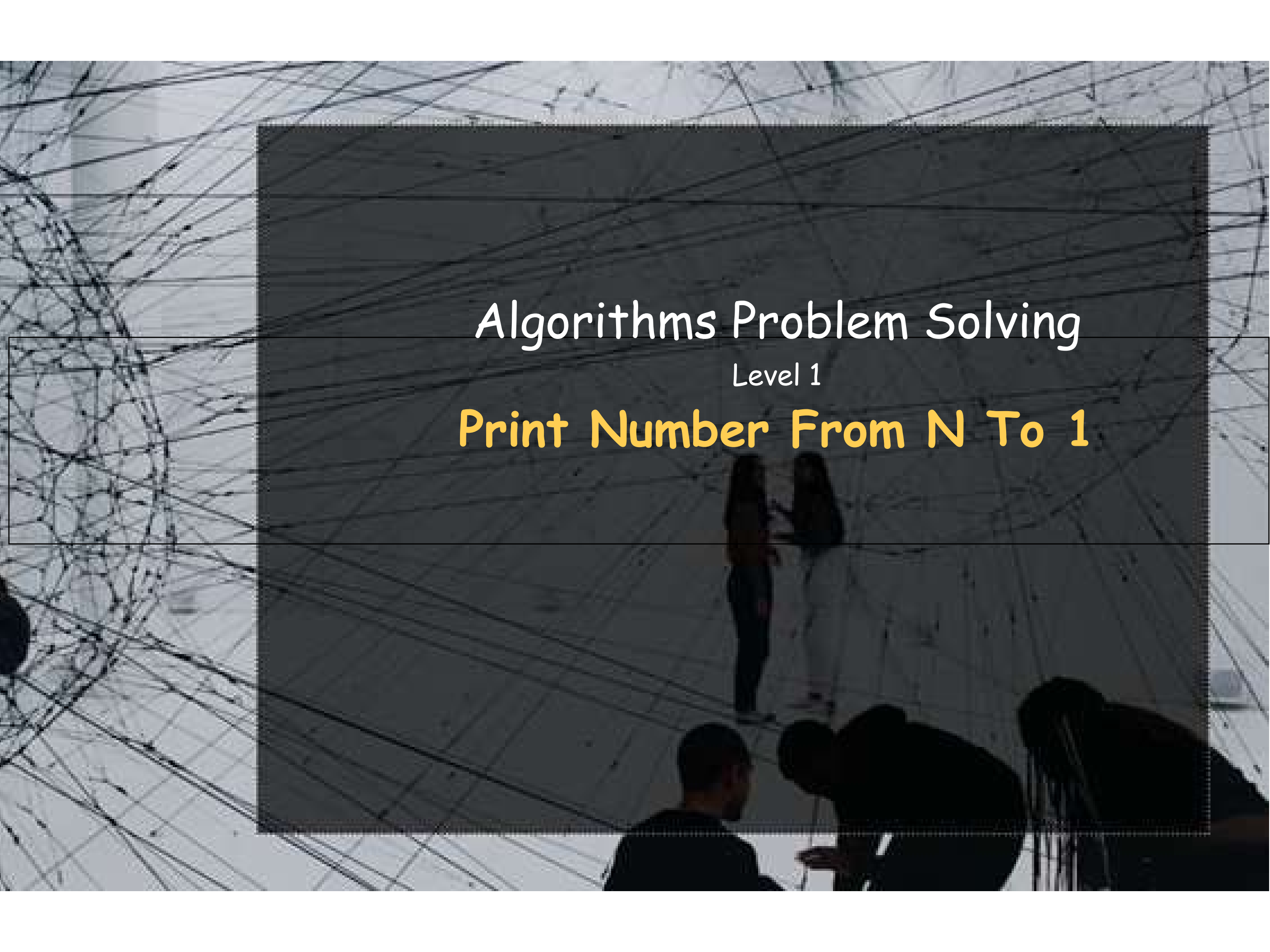


Algorithms Problem Solving

Level 1

Print Number From N To
1



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Print Number From N To 1

Definition

- ◆ This program reads a number N from the user and prints all numbers from N down to 1.
- ◆ It uses a `for` loop with a decreasing counter.
- ◆ The program separates input and output into different functions.

Explanation

- ◆ ``readNumb()`` asks the user to enter a number and returns it.
- ◆ ``printRangeFrom1ToN()`` receives N and starts the counter at N.
- ◆ ``counter >= 1`` keeps the loop running while the counter is at least 1.
- ◆ ``counter--`` decreases the counter by 1 after each iteration.
- ◆ ``printRangeFrom1ToN(readNumb())`` first reads the number, then passes it directly to the printing function.
- ◆ Although the function is named ``printRangeFrom1ToN``, it actually prints from N to 1. A clearer name would be ``printRangeFromNTo1``.

Steps in main()

1. Read the number N from the user.
2. Set `counter` to N.
3. Check whether `counter >= 1`.
4. Print the current value of `counter`.
5. Decrease `counter` by 1.
6. Repeat until `counter` becomes less than 1.

💡 Example

◆ If the user enters:

$N = 5$

◆ The loop starts at 5 and decreases by 1 until it reaches 1.

➡ Result

Range:

5

4

3

2

1

Note

- ◆ ``counter--`` is equivalent to ``counter = counter - 1``.
- ◆ The condition ``counter >= 1`` includes 1 in the output.

⊘ Warning

◆ If the user enters 0 or a negative number, the loop will not execute because the initial value does not satisfy `counter >= 1`.



Comparison

▨ From 1 To N

- ✓ Counter starts at 1.
- ✓ Counter increases using `counter++`.
- ✓ Condition is `counter <= N`.

▤ From N To 1

- ✓ Counter starts at N.
- ✓ Counter decreases using `counter--`.
- ✓ Condition is `counter >= 1`.

✦ Summary

- ✓ The program prints numbers from N down to 1.
- ✓ The counter starts at N.
- ✓ `counter--` decreases the counter by 1.
- ✓ `counter >= 1` keeps the loop running until 1.
- ✓ The function name could be improved to `printRangeFromNTo1`.

Thank you 😊